

Cross-Lingual Transfer through Phonetic Representations: A Multimodal Approach for Low-Resource Intent Classification

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Abstract

Most language models today are trained on text using word or subword tokenization. But getting large amounts of text is difficult and costly, and many low-resource languages simply do not have it. What some of these languages do have is speech. We investigate whether pairing a high-resource language with a related low-resource language, and converting both text and speech to a shared phonetic representation, allows cross-lingual transfer between them. We use French as the high-resource language and evaluate on French-Italian and French-Haitian Creole. We pre-train a phone encoder using MLM on French data along with target-language speech, then fine-tune for a 10-label intent classification task with 400 training examples, extracted from the MASSIVE 1.1 dataset. We test two pipelines: TTS+Allosaurus, which routes both modalities through the same phone recognizer, and the Epitran+Allosaurus pipeline used by [Leong and Whitenack \(2022\)](#). We also compare a from-scratch transformer with a CamemBERT-initialized encoder. MLM pretraining helps most when input data is noisy and the encoder starts from scratch, and helps very little on cleaner phoneme data. CamemBERT initialization gives a large boost over from-scratch training but still trails raw-text fine-tuning, which quantifies the cost of operating in a phone space.

1 Introduction

There are over 7000 spoken languages in the world, but modern NLP systems work well for only a small fraction of them. The reason is uneven data: high-resource languages have large text corpora, while many low-resource languages do not. However, many languages that are short on text have substantial speech data available. This raises a natural question: can we use speech data in a low-resource language together with text data in a re-

lated high-resource language to train shared models?

The challenge is that speech and text live in different representation spaces. Text is a sequence of words or subword tokens, while speech is a continuous audio signal. To train a single model on both, we need a shared representation that both modalities can be mapped into. Phones provide such a representation. They describe distinct sounds and can be produced from speech using a phone recognizer, or from text using a text-to-speech system followed by the same recognizer. Once both modalities are converted to phones, a model can be trained on the combined data.

[Leong and Whitenack \(2022\)](#) explored this idea for named entity recognition in Swahili and Kinyarwanda, using Epitran¹ to convert text to phonemes and Allosaurus² to convert speech to phonemes in the same inventory. They show that MLM pretraining on phone-converted data improves downstream performance.

In this work we build on their approach and contribute the following:

(a) We apply the phone-based transfer setup to a new task, intent classification on the MASSIVE 1.1³ dataset, and a new task family, sentence-level classification rather than sequence labeling.

(b) We propose an alternative pipeline that uses TTS followed by Allosaurus for text input rather than Epitran. Routing both text and speech through the same recognizer keeps the symbol distribution consistent across modalities, and the pipeline does not depend on Allosaurus having a phone inventory available for the target language. This matters because Allosaurus does not support Haitian Creole, so the original Epitran+Allosaurus pipeline cannot be applied there.

¹<https://github.com/dmort27/epitran>

²<https://github.com/xinjli/allosaurus>

³<https://huggingface.co/datasets/AmazonScience/massive>

(c) We compare a small transformer trained from scratch with a CamemBERT-initialized encoder where only the input embedding and output projection are swapped for the phone vocabulary. This isolates the value of starting from a strong pre-trained French model.

(d) We evaluate on two language pairs, French–Italian and French–Haitian Creole, which differ in their relationship to French. Italian is closely related to French through shared Romance origins, while Haitian Creole is a French-derived creole with substantial divergence in grammar and vocabulary.

2 Related Work

Phonetic representations for cross-lingual NLP. Leong and Whitenack (2022) is the closest prior work to ours. They propose phone-level pretraining as a way to bridge text and speech for low-resource NLP, evaluate on Swahili and Kinyarwanda named entity recognition, and find that pretraining on phone-converted data improves downstream performance by up to 6% in F1 on full sequence labeling. Our work applies the same general idea to a different task (intent classification) and proposes an alternative pipeline (TTS+Allosaurus) that removes the requirement for a target-language phone inventory.

Dalmia et al. (2019) earlier showed that phoneme-level language models can be shared multilingually for low-resource speech recognition, and that pretraining helps especially when adapting to a new language with little data. Their work targets ASR rather than downstream NLP tasks, but the general finding that phone-level pretraining transfers across languages is consistent with what we observe.

Phonemes as an additional modality. Nguyen et al. (2023) take a different approach: rather than replacing orthography with phones, they integrate phonemic data as a second input modality alongside text in a multilingual transformer. Their target is cross-lingual transfer among Chinese, Japanese, Korean, and Vietnamese, where script differences hurt orthographic transfer. Their setup keeps the orthographic information and adds phonemes; ours collapses to phones only, which is necessary in our setting because the speech modality has no orthography to start with.

Cross-lingual transfer to low-resource lan-

guages. Chen et al. (2025) address cross-lingual multimodal sentiment analysis using language family disentanglement. Their setting differs from ours (image-text rather than speech-text), but the broader problem is the same: transferring NLP capability from a high-resource language to a low-resource one when modalities differ. Their results suggest that explicit handling of language relatedness can improve transfer, which connects to our observation that French–Italian transfer behaves differently from French–Haitian transfer.

Phonemic resources and tools. Goriely and Buttery (2025) introduce G2P+, a tool that aligns G2P output with established phoneme inventories from the Phoible database, and IPA CHILDES, a multilingual phonemic dataset. Their analysis shows that different G2P tools produce noticeably different phoneme inventories for the same language, which has direct implications for any pipeline that combines G2P with phone recognition. We use this as motivation in our discussion of why the Epitran+Allosaurus pipeline can show inconsistent pretraining gains.

Architecture. Our from-scratch encoder is a standard transformer (Vaswani et al., 2017). The CamemBERT vocab-swap encoder builds on CamemBERT (Martin et al., 2020), a French-only RoBERTa model. CamemBERT being a French-only model helps us validate that cross-lingual transfer is possible for French-Italian and French-Haitian Creole language pairs.

3 Methodology

Our approach maps text and speech data into a shared representation of phones, pretrains an encoder on that data, and then fine-tunes the encoder for intent classification. The high-resource language is French in both of our experiments. The low-resource targets are Italian and Haitian Creole. We compare two methods for producing phonetic representations, and use both to run masked language modeling pretraining followed by fine-tuning, comparing performance and the gain from pretraining.

3.1 Phonetic Pipeline

We compare two pipelines for converting input data into a shared representation. The two pipelines differ in what phonetic data they produce based on how they handle text input.

TTS+Allosaurus (phones). For text input, we first synthesize speech using a TTS model and then run the generated audio through Allosaurus, a universal phone recognizer. For speech input, we run Allosaurus directly. Both modalities pass through the same recognizer, so they end up with the same kind of unit. We do not use Allosaurus’s phone inventory specialization, so the output is a sequence of phones drawn from the universal phone set. Phones represent distinct sounds rather than language-specific contrasts.

We use two TTS models across our experiments. Coqui TTS⁴ was the only model we found that supported Haitian Creole, so we used it for both Haitian Creole and French. For Italian, Coqui did not have a good model available, so we used Piper TTS⁵, which does. This led us to a deliberate choice in selecting Italian as our second low-resource target. Italian is not a low-resource language in general, but using it lets us isolate the effect of the phone-based transfer setup from the effect of TTS quality. We expected the Haitian TTS to be lower quality, so pairing the experiments with Italian — where TTS quality is high — helps separate what is happening because of the language pair from what is happening because of the data pipeline.

Epitran+Allosaurus (phonemes). This is the pipeline used by [Leong and Whitenack \(2022\)](#). For text, we use Epitran, a rule-based grapheme-to-phoneme tool that maps text directly to a language-specific phoneme inventory. For speech, we run Allosaurus with its phone inventory restricted to match Epitran’s output for the target language. Phonemes are language-specific units representing meaningful sound contrasts within a language.

The Epitran+Allosaurus pipeline only works when Allosaurus has a phone inventory available for the target language. Italian is supported, but Haitian Creole is not, so we can only run this pipeline on the French–Italian pair.

We chose to develop the TTS+Allosaurus pipeline rather than relying on Epitran for two reasons. First, Epitran and Allosaurus produce different symbol sequences for the same input even when their inventories overlap, which creates a mismatch between text-derived and speech-derived data. Routing both modalities through Allosaurus avoids this. Second, the TTS+Allosaurus pipeline

does not depend on inventory support, so it extends to languages like Haitian Creole that Allosaurus does not directly support.

3.2 Phone Encoder Architecture

We use two encoders. The first is a small transformer trained from scratch. The second reuses the pretrained weights of CamemBERT⁶ ([Martin et al., 2020](#)), a French-only RoBERTa model. Both produce a sequence representation that is collapsed into a single vector and passed to a linear classifier.

From-scratch encoder. The input is a sequence of phone or phoneme tokens, depending on the pipeline. We use a learned embedding of size 128 with sinusoidal positional encoding ([Vaswani et al., 2017](#)), followed by 4 transformer encoder layers with 4 attention heads, a feedforward dimension of 512. For classification, we apply masked mean pooling over the non-padding positions and pass the result through a linear classifier head. The vocabulary is built from whitespace-split tokens in the pretraining and fine-tuning data, plus standard special tokens (<pad>, <unk>, <mask>, <cls>, <sep>).

CamemBERT vocab-swap. The motivation here is that CamemBERT already encodes a lot of French knowledge, and we would rather adapt that knowledge to phone input than start from scratch. We load CamembertForMaskedLM and call `resize_token_embeddings` with our phone vocabulary size. This swaps two things: the input embedding rows and the MLM output projection rows. Everything else stays as-is, including the transformer layers and the MLM head’s intermediate dense layer. Only the parts that have to change for the new vocabulary are reinitialized. For classification, we take the encoder from the MLM model and add a fresh classification head: dropout, masked mean pooling, and a linear layer. CamemBERT does not provide a pretrained classification head for this task, so this part is always trained from scratch.

3.3 MLM Pretraining

Both encoders are pretrained with masked language modeling on phone or phoneme sequences. We use standard BERT-style masking with a probability of 0.15 and the 80/10/10 split: 80% of selected positions are replaced with <mask>, 10% with a random token, and 10% are kept unchanged. The

⁴<https://github.com/coqui-ai/tts>

⁵<https://github.com/rhasspy/piper>

⁶https://huggingface.co/docs/transformers/model_doc/camembert

loss is cross-entropy over the masked positions only.

3.4 CamemBERT Continued Pretraining

When MLM pretraining is applied to the CamemBERT vocab-swap encoder, the setup is the same as above but with one important difference: the transformer layers start from CamemBERT’s pre-trained weights rather than from random initialization. The MLM objective therefore acts as continued pretraining, gradually adapting CamemBERT’s French representations to operate over phone sequences instead of subword tokens.

The input embedding and output projection rows are randomly initialized at the start (since the phone vocabulary is new), but they receive gradient updates throughout pretraining.

3.5 Fine-tuning

We fine-tune on the MASSIVE 1.1 dataset, which contains intent-labeled utterances across many languages. For phone-level models, the input utterances are passed through the same pipeline used during pretraining. For text-only baselines (Section 4.2), the input is the raw text in the target language.

For phone-level models we report the average test accuracy over the last 5 fine-tuning epochs. We do this because the small encoder can be unstable in the final epochs, and averaging gives a more honest picture than picking the best epoch. For the text-only baselines we report the best test accuracy across epochs, because these larger models train stably and the best-epoch number is a reasonable summary. We acknowledge this difference in metric and note that it makes our comparison more conservative: we compare our average against their best.

4 Experiments

4.1 Datasets

We use three sources of data.

Pretraining text. We sample 100k French sequences from the mC4⁷ corpus. This is passed through the appropriate phonetic pipeline to produce phone or phoneme sequences for MLM pretraining.

Pretraining speech. We use 10–20 hours of speech in the target language drawn. For Haitian

Creole we used data published by Carnegie Mellon⁸. While for Italian we used data from Mozilla Common Voice⁹ This is processed by Allosaurus directly, since the input is already audio. Including target-language speech during pretraining exposes the model to the phone distributions it will encounter at fine-tuning time.

Fine-tuning. We use the MASSIVE 1.1 dataset for intent classification. MASSIVE is a multilingual intent classification dataset with utterances labeled across many languages. We restrict to a 10-class subset of intents and use 400 training examples and 100 test examples per language. The 400/100 split is chosen to simulate the low-resource conditions our setup is designed for.

4.2 Baselines

We report several text-only baselines that operate directly on raw text in the target language. These help quantify what accuracy is achievable without going through a phone representation, and how much our phone-level models trail by.

LLM baselines (LoRA fine-tuning). We fine-tune three multilingual large language models with LoRA: Llama-3.2-1B, Llama-3.1-8B, and Qwen3-8B. The Llama models cover a small set of mainstream languages including French and Italian. Qwen3-8B was trained on a much broader set of around 200 languages, which includes all three of our target languages. For each model we report the best accuracy across either 15 or 20 fine-tuning epochs, whichever produced a higher number for that model.

CamemBERT (raw text). We fine-tune CamemBERT directly on the raw text of the target language using CamembertForSequenceClassification. This is full fine-tuning with CamemBERT’s own subword tokenizer. Because CamemBERT only knows French, this baseline measures how much French-to-related-language transfer is happening at the text level. It is a particularly informative comparison for our CamemBERT vocab-swap encoder, since the only structural difference is whether the input is text or phones.

4.3 Implementation Details

Hyperparameters used in each setup are listed in Table 3. All phone-level models are trained with

⁷https://huggingface.co/datasets/allenai/c4/tree/mC4_3.1.0/multilingual

⁸<http://www.speech.cs.cmu.edu/haitian/>

⁹<https://mozilladatacollective.com/datasets/cmn2h0yei01msmm07u8z5vu87>

AdamW. The CamemBERT vocab-swap setup uses standard BERT-style 80/10/10 masking with probability 0.15. LoRA fine-tuning uses rank 16 and a learning rate of $1e-4$. We do not use early stopping in any of our runs; instead, we report the average of the last 5 epochs for phone-level models and the best epoch for text baselines, as discussed above.

5 Results

5.1 Italian Results

Table 1 shows results for the French–Italian pair. Several patterns stand out.

For the from-scratch encoder, MLM pretraining helps substantially on TTS+Allosaurus phones, raising accuracy from 0.34 to 0.41. On the Epitran+Allosaurus phonemes used by [Leong and Whitenack \(2022\)](#), the same pretraining setup gives only a marginal gain, from 0.45 to 0.46. The two pipelines respond very differently to the same pretraining objective, which we discuss in Section 6.

The phone-based CamemBERT encoder is much stronger than the from-scratch encoder across the board. With Epitran+Allosaurus phonemes and pretraining, it reaches 0.64, while the equivalent from-scratch result is 0.46. With TTS+Allosaurus phones it reaches 0.48, also a clear improvement over the from-scratch result of 0.43. The pretraining gain is smaller than for the from-scratch encoder, suggesting that some of what MLM learns from scratch is already present in CamemBERT’s pretrained weights.

The text-based baselines establish an upper bound. CamemBERT fine-tuned on raw Italian text reaches 0.79, and Qwen3-8B reaches 0.80.

5.2 Haitian Creole Results

Table 2 shows results for the French–Haitian Creole pair. The matrix is smaller because the Epitran+Allosaurus pipeline is unavailable for Haitian Creole.

The same overall pattern holds: MLM pretraining helps the from-scratch encoder on TTS+Allosaurus phones, raising accuracy from 0.27 to 0.32, and the phone-based CamemBERT encoder outperforms from-scratch, reaching 0.38 with pretraining versus 0.32. Pretraining helps the phone-based CamemBERT encoder as well, raising accuracy from 0.32 to 0.38.

Absolute scores are noticeably lower than for Italian. Several factors likely contribute: TTS quality is harder to ensure for a low-resource language

like Haitian Creole, the linguistic distance from French is larger than for Italian, and the speech pretraining data may be of lower quality. The text-based baselines also drop relative to Italian (CamemBERT raw text falls from 0.79 to 0.77, Qwen3-8B from 0.80 to 0.66), confirming that some of the difficulty is at the language level rather than at the phone-pipeline level.

The CamemBERT raw-text baseline at 0.77 on Haitian Creole is striking. CamemBERT only knows French, and yet it competes with much larger multilingual models on Haitian Creole text. This suggests that French-to-Haitian transfer at the text level is strong.

5.3 Comparison with Prior Work

A direct head-to-head comparison with [Leong and Whitenack \(2022\)](#) is not possible because of differences in task (intent classification vs. NER), metric (accuracy vs. F1), and language pairs (French–Italian/Haitian vs. Swahili/Kinyarwanda). What we can compare is the rough scale of the pretraining effect.

They report up to 6% improvement in F1 on their hardest NER task, which is full sequence labeling. For our TTS+Allosaurus pipeline, pretraining raises accuracy from 0.34 to 0.41 on French–Italian and from 0.27 to 0.32 on French–Haitian, which is in the same general range. However, their pipeline (Epitran+Allosaurus with inventory constraints) shows almost no pretraining gain in our setup, with accuracy moving only from 0.45 to 0.46 on Italian. The gain we observe comes specifically from our TTS+Allosaurus pipeline. We discuss possible reasons in the next section.

6 Discussion

6.1 When Does Pretraining Help?

The clearest pattern in our results is that MLM pretraining helps a lot on TTS+Allosaurus phones, raising accuracy from 0.34 to 0.41 on Italian and from 0.27 to 0.32 on Haitian, but helps very little on Epitran+Allosaurus phonemes, where accuracy moves only from 0.45 to 0.46 on Italian. This holds for both encoder architectures.

One explanation is noise. TTS+Allosaurus produces phones by passing text through a TTS model and then a phone recognizer, both of which introduce errors. Epitran is deterministic: a given input always produces the same output. MLM pretraining teaches the model to predict masked tokens

Model	Input Representation	No Pretrain	With MLM Pretrain
<i>Phone-level models</i>			
From-scratch encoder	TTS+Allosaurus phones	0.34	0.41
From-scratch encoder	Epitrans+Allosaurus phonemes	0.45	0.46
CamemBERT (vocab-swap)	TTS+Allosaurus phones	0.43	0.48
CamemBERT (vocab-swap)	Epitrans+Allosaurus phonemes	0.61	0.64
<i>Text-based baselines (fine-tuned only, no additional MLM)</i>			
Llama-3.2-1B (LoRA)	Raw text (Italian)		0.72
Llama-3.1-8B (LoRA)	Raw text (Italian)		0.71
Qwen3-8B (LoRA)	Raw text (Italian)		0.80
CamemBERT (full FT)	Raw text (Italian)		0.79

Table 1: Italian intent classification accuracy (avg over last 5 epochs for phone-level models; best epoch for text baselines).

Model	Input Representation	No Pretrain	With MLM Pretrain
<i>Phone-level models</i>			
From-scratch encoder	TTS+Allosaurus phones	0.27	0.32
CamemBERT (vocab-swap)	TTS+Allosaurus phones	0.33	0.38
<i>Text-based baselines (fine-tuned only, no additional MLM)</i>			
Llama-3.2-1B (LoRA)	Raw text (Haitian)		0.59
Llama-3.1-8B (LoRA)	Raw text (Haitian)		0.64
Qwen3-8B (LoRA)	Raw text (Haitian)		0.66
CamemBERT (full FT)	Raw text (Haitian)		0.77

Table 2: Haitian Creole intent classification accuracy (avg over last 5 epochs for phone-level models; best epoch for text baselines). The Epitrans+Allosaurus pipeline is unavailable for Haitian since Allosaurus does not support a Haitian phone inventory.

from context, which is exactly the kind of objective that should help when input data has variability. On clean Epitrans data there is less variability to recover from, so the pretraining objective has less to teach.

This reframes the role of pretraining in this kind of setup. It is not just learning cross-lingual phonotactics; a meaningful part of what it learns is robustness to the noise introduced by the phonetic pipeline itself. This has practical implications: pipelines that produce noisier output may benefit more from pretraining than cleaner pipelines.

6.2 Phones vs. Phonemes: Representation Mismatch

A separate but related issue is that Epitrans and Allosaurus produce different symbol sequences for the same input, even when their phone inventories overlap. We did not investigate this in detail, but observed it consistently while building the data pipelines. This means that a model pretrained on Epitrans-produced phonemes would see a different distribution of symbols than what Allosaurus produces from speech, even with the same target in-

ventory.

Our TTS+Allosaurus pipeline avoids this issue by routing both modalities through the same recognizer. Whatever errors Allosaurus makes, it makes them consistently for both text-derived and speech-derived input. This may be part of why the pretraining gain is larger for our pipeline.

6.3 The Cost of Phone Representations

Our best phone-level result on Italian is 0.64 (phone-based CamemBERT with Epitrans+Allosaurus phonemes and pretraining). The corresponding text-only baseline is CamemBERT fine-tuned on raw Italian text, which reaches 0.79. The gap of 15 points is the cost of working in a phone space.

The trade-off may be worth it in settings where speech is the primary or only available modality. For tasks where text is reliably available, going through phones is unlikely to be the right choice. The interesting case is mixed-modality settings where some data is speech and some is text. Our results suggest the phone-level approach is workable but lossy.

Model	Language Pair	LR	Batch	Pretrain Ep.	Finetune Ep.	Other
<i>Phone-level models</i>						
From-scratch encoder	Fr-It	1e-4	32	—	50	—
From-scratch encoder	Fr-Ht	1e-4	32	—	100	—
From-scratch encoder	Fr-It	5e-5	32	20	30	—
From-scratch encoder	Fr-Ht	5e-5	32	30	30	—
CamemBERT (vocab-swap)	Fr-It	5e-5	32	2	50	—
CamemBERT (vocab-swap)	Fr-Ht	5e-5	32	2	50	—
<i>Text-based baselines (fine-tuned only)</i>						
Llama-3.2-1B (LoRA)	Fr-It	1e-4	16	—	20	rank=16
Llama-3.2-1B (LoRA)	Fr-Ht	1e-4	16	—	20	rank=16
Llama-3.1-8B (LoRA)	Fr-It	1e-4	16	—	15	rank=16
Llama-3.1-8B (LoRA)	Fr-Ht	1e-4	16	—	20	rank=16
Qwen3-8B (LoRA)	Fr-It	1e-4	16	—	20	rank=16
Qwen3-8B (LoRA)	Fr-Ht	1e-4	16	—	20	rank=16
CamemBERT (full FT)	Fr-It	2e-5	16	—	50	—
CamemBERT (full FT)	Fr-Ht	2e-5	16	—	50	—

Table 3: Hyperparameters used for each model per language pair. “Pretrain Ep.” applies only to phone-level models with MLM pretraining. rank refers to the lora_rank hyperparameter

6.4 Why CamemBERT Outperforms From-Scratch

The from-scratch encoder has to learn the structure of phonetic representations, how phones map across languages, and the intent classification task all from limited data. The phone-based CamemBERT encoder inherits a transformer that already encodes a lot of French structure; what it needs to learn is mainly how to map the new phone vocabulary into that existing representation space.

The empirical gap between the two is large. On Italian with Epitran+Allosaurus and pretraining, the phone-based CamemBERT encoder reaches 0.64 while the from-scratch encoder reaches 0.46. The gap holds across pipelines and language pairs. This is consistent with the broader finding by Gururangan et al. (2020) that continued pretraining of a strong language model on domain-relevant data outperforms training from scratch, even when the target domain is quite different from the original pretraining domain. Our setup is an extreme version of this: the input vocabulary is replaced entirely with a phone inventory, and yet the pretrained French knowledge in CamemBERT still transfers usefully. This suggests that phone-based work in this setting should generally start from a strong pretrained model in a related language rather than training from scratch.

6.5 Limitations

Our work has several limitations worth flagging.

TTS quality is a real bottleneck. We used Piper TTS, which produces good French and Italian au-

dio but for Haitian Creole finding a good quality TTS model is challenging. Some of the gap between Italian and Haitian Creole results may be attributable to TTS quality rather than to anything intrinsic about Haitian Creole as a language.

Allosaurus has its own non-trivial phone error rate, reported by Leong and Whitenack (2022) at around 24%. We did not measure phone error rate directly in our setup. A noisier phone recognizer means a noisier representation throughout, which affects everything downstream.

We use only 400 training examples and 100 test examples per language. This simulates low-resource conditions, but conclusions drawn from such small samples should be treated with caution. The gap between phone-level models and text baselines may shrink with more data, or it may not.

We tested only two language pairs, and both target languages are in some way related to French (Italian through shared Romance origins, Haitian Creole through derivation). The behavior of this approach on language pairs that are not related is an open question.

Finally, our discussion of why pretraining helps differently across pipelines is a hypothesis based on observed patterns. We do not have direct measurements of input noise across pipelines, and disentangling noise robustness from cross-lingual transfer would require additional experiments.

7 Conclusion

We presented a phone-based approach to cross-lingual transfer for intent classification in low-

resource languages. Both text and speech are converted to phones using a shared recognizer, and a model is trained on the combined data. We evaluated on two language pairs, French–Italian and French–Haitian Creole, and compared our TTS+Allosaurus pipeline with the Epitran+Allosaurus pipeline of [Leong and Whitenack \(2022\)](#), and a from-scratch encoder with a phone-based CamemBERT encoder.

MLM pretraining helps most when the phone pipeline is noisy and the encoder starts from scratch. When the encoder starts from CamemBERT or the pipeline is cleaner, pretraining helps less. The phone-based CamemBERT setup outperforms from-scratch training across all conditions, recovering most but not all of the gap to text-based baselines. On Italian, the best phone-level result is 0.64 while CamemBERT fine-tuned on raw text reaches 0.79. The gap between phone-level and raw-text models is the cost of being able to handle speech input.

Future work could explore G2P+ ([Goriely and Buttery, 2025](#)) as a cleaner alternative to Epitran for text-to-phoneme conversion, since aligning G2P output with established phoneme inventories may reduce the symbol-level mismatch we observed. Using real low-resource speech rather than TTS-generated speech for pretraining would test whether TTS quality is a major bottleneck, especially for Haitian Creole, where TTS quality is harder to ensure. Adding more language pairs, especially pairs that are not related to French, would help disentangle the noise-robustness and cross-lingual-transfer effects of pretraining

8 Code and Data

Code and data can be found at https://github.com/OmKurkure/5525_Project

Contributions

Om Kurkure tested TTS models and collected the French–Haitian Creole data, and applied the TTS+Allosaurus pipeline to both language pairs. He was responsible for pretraining and fine-tuning the from-scratch encoder and the phone-based CamemBERT encoder on phone and phoneme data for both language pairs. For the report, he wrote the abstract, introduction, methodology, and prepared the results tables. Handled Github setup.

Manu Manivannan Sundarasolai collected the French–Italian data and applied both phonetic

pipelines to it. He was responsible for fine-tuning the Llama, Qwen, and CamemBERT text-only baselines for both language pairs. For the report, he wrote the experiments, results, limitations, and conclusion sections.

The training scripts were written and tested together in person, and the related work and discussion sections were authored jointly.

9 Acknowledgements

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